

Name: Krishna Raviraj Shimpi
PRN No: 202501040036

4.1.1. Pandas - series creation and manipulation

03:12

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the groupby and mean() operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

```
# import pandas as pd

# # Take inputs from the user to create a list of numbers
# numbers = list(map(int, input().split()))

# grouped =

# # Display the mean of even and odd numbers with labels
# grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]
# print("Mean of even and odd numbers:")
# print(grouped)
import pandas as pd

# Take inputs from the user to create a list of numbers
numbers = list(map(int, input().split()))

# Create a Pandas series from the list of numbers
s = pd.Series(numbers)

# Grouping by even and odd numbers and calculating the mean
grouped = s.groupby(s % 2 == 0).mean()

# Display the mean of even and odd numbers with labels
grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]
```

```
print("Mean of even and odd numbers:")
```

```
print(grouped)
```

Explorer Debugger

Average time: 0.007 s (7.00 ms) Maximum time: 0.017 s (17.00 ms)

3 out of 3 shown test case(s) passed
3 out of 3 hidden test case(s) passed

Test case 1 (17 ms) [Debug]

Expected output	Actual output
1 2 3 4 5 6 7 8 9 10	1 2 3 4 5 6 7 8 9 10
Mean of even and odd numbers:	Mean of even and odd numbers:
Odd: 5.0	Odd: 5.0
Even: 6.0	Even: 6.0
dtype: float64	dtype: float64

Test case 2 (5 ms)

Test case 3 (5 ms)